

Black-Box Optimization for IFC Power Control via RIBBO

Progress Report — May 2026

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Setup

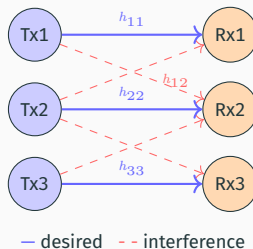
(Lee et al., TCOM 2025, Sec. V-A)

- $D=3$ transmitter-receiver pairs
- Action: $\mathbf{x} \in [0, 1]^3$ (per-pair power ratio)
- Channel: Rayleigh fading, $h_{d'd} \sim \mathcal{CN}(0, 1)$
- $P_{\text{tx}} = 10 \text{ W}$

Objective — Spectral Efficiency (SE):

$$r_{\text{SE}}(\mathbf{x}) = \sum_{d=1}^D \log \left(1 + \frac{P_{\text{tx}} |h_{dd}|^2 x_d}{1 + P_{\text{tx}} \sum_{d' \neq d} |h_{d'd}|^2 x_{d'}} \right)$$

Goal: maximize r_{SE} for each new channel realization

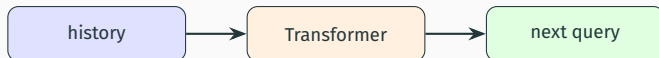


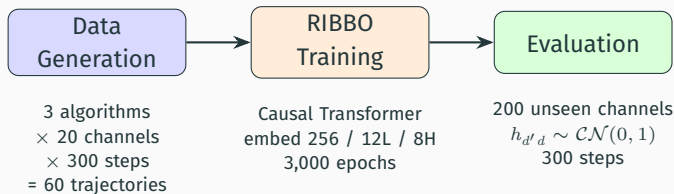
Core idea

- Learn a BBO *algorithm* (not a solution)
- Input: history ($\mathbf{x}_{1:t}, y_{1:t}, R_{1:t}$)
- Output: next query $\hat{\mathbf{x}}_{t+1}$
- Backbone: Causal Transformer (GPT-2)

Key ingredients

- **RTG** (Regret-To-Go): conditions model on remaining improvement budget
- **HRR** (Hindsight Regret Relabelling): relabels any offline trajectory as near-optimal
- **Data**: K algorithms $\times N$ tasks $\times T$ steps





Behavior algorithms (data gen)

- Random Search
- Hill Climbing
- CMA-ES

Key implementation

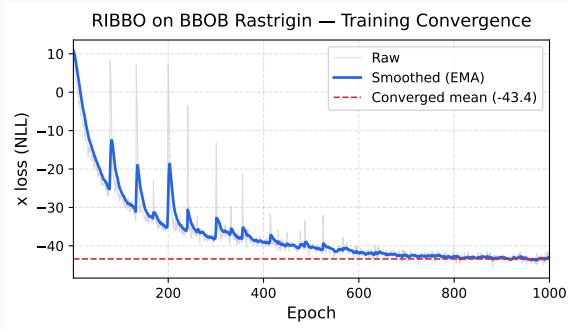
- IFCMetaProblem: IFC task class added to RIBBO codebase
- Per-channel RTG normalization
- Checkpoint resume across training sessions

Setup

- Problem: Rastrigin (dim=10, BBOB)
- 20 tasks $\times$ 3 algorithms $\times$ 300 steps
- 1,000 epochs, RTX 4070 Ti ($\approx$ 51 min)

Result

- x-loss (Gaussian NLL): converges by epoch $\approx$ 600
- Early oscillation due to LR warmup (10k steps)
- Epoch 1: +10.8 $\rightarrow$ Epoch 1000: -44.9

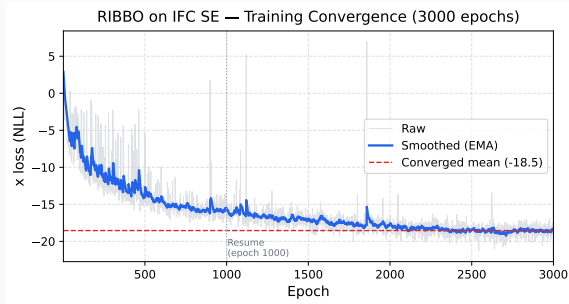


Setup

- 20 train channels $\times$ 3 algorithms $\times$ 300 steps
- embed 256 / 12 layers / 8 heads
- RTX 4070 Ti, ≈ 2.5 h

Result

- Epoch 1: +2.9 $\rightarrow$ Epoch 3000: -18.6
- Oscillation larger than Rastrigin (channel-specific reward scale)
- Stabilizes by epoch $\approx 2,500$



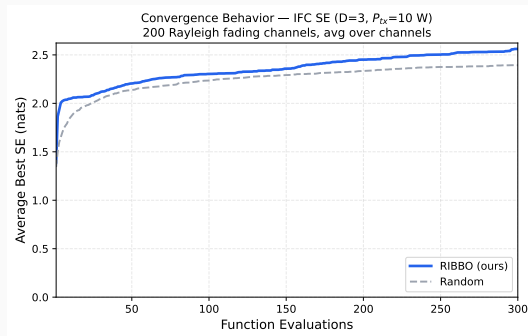
Evaluation setup

- 200 unseen channels,
 $h_{d'd} \sim \mathcal{CN}(0, 1)$
- 300 steps (in-distribution)
- Metric: best r_{SE} found so far (avg over channels)

Final result @ step 300

Method	nats	bits
RIBBO (ours)	2.67	3.85
Random	2.44	3.52

RIBBO +9.5% over random baseline



Completed

- ✓ IFC SE task integrated into RIBBO codebase
- ✓ 60 offline trajectories generated (3 algos $\times$ 20 channels)
- ✓ Training converged: 3,000 epochs
- ✓ RIBBO outperforms random on 200 unseen channels

Next steps

- Compare against CMA-ES baseline (strongest behavior algo)
- Extend trajectory length to 500 steps for direct comparison with Lee et al.
- EE reward variant